

**YEAR 9 SCIENCE
CHEMICALS SCIENCES
TEST 2 - CHEMICAL REACTIONS**

NAME: SOLUTIONS

CLASS: _____

Mark: 33

Chemical reactions, including combustion and the reactions of acids, are important in both non-living and living systems and involve energy transfer.

Multiple Choice Write the answer to each question in the appropriate box at right.

1. Which of the following word equations correctly explains an acid reacting with a metal?

- (a) zinc + hydrochloric acid → zinc chloride + hydrogen
- (b) magnesium + sulphuric acid → magnesium chloride + hydrogen
- (c) magnesium + hydrochloric acid → carbon dioxide + hydrogen
- (d) zinc + hydrochloric acid → zinc chloride + carbon dioxide

2. Which of the following word equations correctly explains an acid reacting with a carbonate?

- (a) hydrochloric acid + calcium carbonate → calcium chloride + water + carbon dioxide
- (b) hydrochloric acid + calcium carbonate → calcium chloride + water + hydrogen
- (c) sulphuric acid + calcium carbonate → calcium chloride + water + carbon dioxide
- (d) sulphuric acid + calcium carbonate → calcium sulphate + hydrogen + carbon dioxide

3. Which of the following is **not** a property of acids?

- (a) React with some metals to produce hydrogen gas.
- (b) Turn litmus red.
- (c) Sweet taste.
- (d) Sour taste.

4. Which of the following is a **base**?

- (a) NaCl (sodium chloride)
- (b) NaOH (sodium hydroxide)
- (c) CH₃OH (methanol)
- (d) HCl (hydrochloric acid)

5. Which of the following could be the pH for a pool that had turned **slightly acidic**?

- (a) 2
- (b) 6.5
- (c) 7.5
- (d) 7

QUESTION	ANSWER
1	A
2	A
3	C
4	B
5	B
6	A
7	A
8	A
9	A
10	B
11	B
12	C

6. Which of the following is the correct **general equation** to describe the reaction of an acid and a base?
- (a) acid + base $\rightarrow$ a salt + water
 - (b) acid + base $\rightarrow$ metal hydroxide + carbon dioxide
 - (c) acid + base $\rightarrow$ carbon dioxide + water
 - (d) acid + base $\rightarrow$ hydrogen + water
7. Magnesium hydroxide can be taken to settle stomach acids. The reason why it is useful for this is because:
- (a) the magnesium hydroxide neutralises the stomach acid.
 - (b) the magnesium hydroxide is not very acidic so it won't make your stomach more acidic.
 - (c) magnesium hydroxide is a base so it will turn the stomach acid into a salt.
 - (d) stomach acid will turn into a base when the magnesium hydroxide is taken.
8. Which of the following is most likely to be the pH of lemon juice?
- (a) 3
 - (b) 7
 - (c) 10
 - (d) 14
9. Which of the following will turn universal indicator **red**?
- (a) Vinegar.
 - (b) Sodium hydroxide.
 - (c) Distilled water.
 - (d) Glucose solution.
10. A chemist tells you that a substance she has tested has a pH of 1. From this you know:
- (a) its colour.
 - (b) that it is a strong acid.
 - (c) that it is a weak base.
 - (d) that it is concentrated.
11. Which of the following facts about toothpaste explain why it helps reduce tooth decay?
- (a) It has a minty taste.
 - (b) It is slightly basic.
 - (c) It contains fluoride.
 - (d) It is abrasive.
12. The test for determining whether a colourless, odourless gas is hydrogen is called the:
- (a) glowing splint test.
 - (b) pH test.
 - (c) pop test.
 - (d) limewater test.

Short Answer

Write the answer to the questions in the spaces provided.

1. List **two** characteristics of acids and bases. (Any 2 for each - 1 mark each.)

Acids

- Produce H^+ ions in solution.
- Have a sour taste.
- Turn blue litmus red.

Bases

- Produce OH^- ions in solution.
- Any metal oxide or hydroxide.
- Have a soapy feel.
- Have a bitter taste.
- Turn red litmus blue.

(4)

2. Classify the following compounds as **ACID**, **BASE** or **SALT**.

H_2SO_4 $Mg(OH)_2$ sodium chloride Al_2O_3 HNO_3 $CuSO_4$ calcium oxide HCl

ACID	BASE	SALT
H_2SO_4 HNO_3 HCl	$Mg(OH)_2$ Al_2O_3 calcium oxide (½ mark off each error.)	sodium chloride $CuSO_4$

(4)

3. Complete the following table, giving **estimates on the pH where necessary**, the **colour of the universal indicator** and whether the substance is a **strong/weak acid**, a **strong/weak base** or **neutral**.

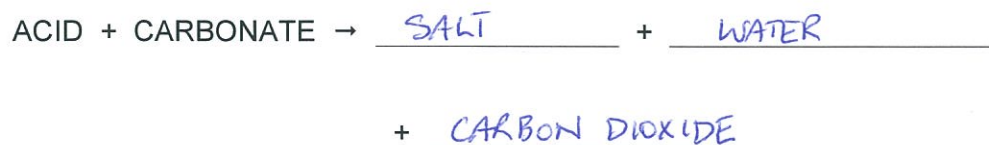
SUBSTANCE	pH	UNIVERSAL INDICATOR (colour)	ACID / BASE/ NEUTRAL (strong / weak)
vinegar	5	orange / yellow	weak acid
soda water	3	light red	weak acid
2.00 M H_2SO_4 (sulphuric acid)	0/1	red	strong acid
1.00 M Na_2SO_4 (sodium sulphate)	8	light blue	weak base
sugar solution	7	green	neutral
yoghurt	5/6	yellow	weak acid

↑
accept either.

(½ mark off each error.)

(6)

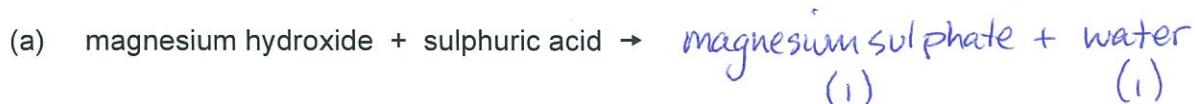
4. Complete the following general word equations for acid reactions.



($\frac{1}{2}$ mark off each error.)

(3)

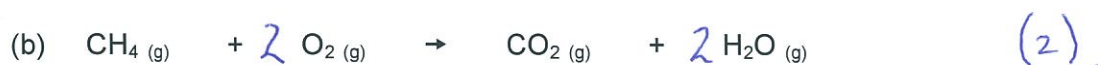
5. Complete the following **WORD EQUATIONS** for acid reactions, naming the salt produced and other products.



(4)

6. **Optional**

Balance the following equations.



(4)